PDE-RSI Benchmark Problems

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1 Protocols, data and scoring

Student protocol. For the `pde_benchmarks.tex` suites and the inventory suite, an agent receives, for each case, 256 noisy samples of the solution (Gaussian noise with standard deviation 1 % of the samples' RMS, per component), the domain, and black-box forcing, initial, boundary and (where present) coefficient data. It must identify the equation class and its coefficients from a library of candidate equations and return the solution over the whole domain. Case identifiers are hash-ordered and reveal neither the class nor the problem.

Code protocol. For the manufactured suites (Section 4) the equation is given and the agent writes one solver per family.

Scoring. The returned solution is linearly interpolated onto a uniform evaluation grid and compared with the exact or a high-accuracy reference solution using the trapezoidal rule. A case passes when the space–time and the final-time relative L^2 errors are both at most the tolerance, the initial and boundary conditions hold to 10^{-10}, and the solve fits the time limit (wall and CPU time, including start-up) and the memory limit. The reported class and coefficients are scored separately and never gate a case.

Splits. Training (evolution) cases come with labelled feedback; validation (development) cases drive the acceptance gate; sealed test cases are never shown to any agent and are scored once, after the solver is frozen.

Suite	Cases per split	Tolerance	Limits	Evaluation grid
`pde_benchmarks.tex` 1–11	16 / 16 / 17, 20, 20	2×10^{-3}	90 s, 16 GB	1-D 512×500; 2-D $128^2 \times 100$
`pde_benchmarks.tex` 12–16	10 / 10 / 10	2×10^{-3}	90 s (1-D), 300 s (2-D), 16 GB	1-D 512×501; 2-D $129^2 \times 21$
PDE Data Inventory	12 / 12 / 12, 12	2×10^{-3} (4-D: 5×10^{-2})	90 s / 300 s, 16 GB	1-D 512×500; 2-D 256^2 or $128^2 \times 11$
Manufactured (code protocol)	24 / 24 / 24 (1-D)	2×10^{-3}	15 s, 1 GB	1-D 512×500; wave $128^2 \times 100$

Vector solutions (Problems 13, 15, 16) are scored with all components together; the pressure is compared after removing its spatial mean.

2 `pde_benchmarks.tex`: Problems 1–16

Problems 1–11 form the `pdebench` suite and Problems 12–16 the `pdebench-ext` suite. The definitions below are those of `pde_benchmarks.tex`.

1 Preliminaries

Spatial Profiles

Cases 1a–4a and Problems 5, 7, and 8 are posed on $x \in (0, L)$ (on $(0, L)^2$ for Problem 5) with homogeneous Dirichlet boundary conditions. The remaining cases of Problems 1–4 specify their own domains and boundary data. The spatial profile s is one of the following:

$$s(x) = \begin{cases} \sin(\pi x/L), & \text{training,} \\ 4\,\dfrac{x}{L}\left(1 - \dfrac{x}{L}\right), & \text{validation,} \\ \sin(\pi x/L) + \dfrac{1}{4}\sin(3\pi x/L), & \text{reserved.} \end{cases}$$

Each profile satisfies $s(0) = s(L) = 0$. The derivatives are

$$\begin{aligned}
&\text{training:} && s' = \frac{\pi}{L}\cos\frac{\pi x}{L}, && s'' = -\frac{\pi^2}{L^2}\sin\frac{\pi x}{L},\\
&\text{validation:} && s' = \frac{4}{L}\left(1 - \frac{2x}{L}\right), && s'' = -\frac{8}{L^2},\\
&\text{reserved:} && s' = \frac{\pi}{L}\cos\frac{\pi x}{L} + \frac{3\pi}{4L}\cos\frac{3\pi x}{L}, && s'' = -\frac{\pi^2}{L^2}\sin\frac{\pi x}{L} - \frac{9\pi^2}{4L^2}\sin\frac{3\pi x}{L}.
\end{aligned}$$

Temporal Profile

For Cases 2a–4a and Problems 7 and 8, the exact solution is

$$u(x,t) = A\,s(x)g(t), \qquad g(t) = 1 + t^{\beta}, \qquad (x,t) \in (0,L)\times(0,T],$$

with initial condition $u(x,0) = A\,s(x)$. For $0 < \beta < 1$, the factor $\beta t^{\beta-1}$ appearing in the source terms of Cases 2a–4a is unbounded as $t \to 0^+$; the choice $\beta \geq 1$ yields a bounded source term. Problem 5 uses a different temporal profile, given in its definition.

Caputo Derivative

For $0 < \alpha < 1$, the Caputo derivative of order α with respect to t is

$${}_0^C D_t^{\alpha} u(x,t) = \frac{1}{\Gamma(1-\alpha)}\int_0^t (t-\tau)^{-\alpha}\frac{\partial u(x,\tau)}{\partial \tau}\,d\tau.$$

It satisfies ${}_0^C D_t^{\alpha} 1 = 0$ and, for $p > 0$,

$${}_0^C D_t^{\alpha} t^p = \frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\,t^{p-\alpha}.$$

For a variable order $\alpha = \alpha(x,t)$ with $0 < \alpha(x,t) < 1$, we use the definition in which the order is evaluated at the point (x,t) and held fixed throughout the integral:

$${}_0^C D_t^{\alpha(x,t)} u(x,t) = \frac{1}{\Gamma(1-\alpha(x,t))}\int_0^t (t-\tau)^{-\alpha(x,t)}\frac{\partial u(x,\tau)}{\partial \tau}\,d\tau.$$

Under this definition, the power rule above holds pointwise with α replaced by $\alpha(x,t)$.

2 Elliptic Problems

Problem 1. Poisson Equation

$$\boxed{-D\,u_{xx} = f(x)}$$

Case 1a (manufactured). $x \in (0,L)$, $u(0) = u(L) = 0$.

$$u(x) = A\,s(x), \qquad f(x) = -D\,A\,s''(x).$$

Case 1b (multiscale solution; Wang, Wang, and Perdikaris [15]). $x \in (0,1)$, $u(0) = u(1) = 0$.

$$u(x) = \sin(2\pi x) + 0.1\sin(50\pi x), \qquad f(x) = D\left[4\pi^2 \sin(2\pi x) + 250\pi^2 \sin(50\pi x)\right].$$

In [15] the equation is written as $u_{xx} = f$ with $D = 1$; the source term above corresponds to the form $-D\,u_{xx} = f$. The two components of f have comparable amplitudes ($4\pi^2$ and $250\pi^2$), whereas the high-frequency component of u is ten times smaller than the low-frequency one. The case is used to assess the ability of a method to resolve multiscale solutions.

Case 1c (interior layer). $x \in (0,1)$, with Dirichlet data $u(0)$ and $u(1)$ taken from the exact solution. For $\alpha > 0$ and $x_0 \in (0,1)$,

$$u(x) = \arctan\big(\alpha(x - x_0)\big), \qquad f(x) = \frac{2D\alpha^3(x - x_0)}{\left[1 + \alpha^2(x - x_0)^2\right]^2}.$$

This is the one-dimensional analogue of the "wave front" problem in Mitchell's collection of elliptic test problems [8], where the solution is $\arctan(\alpha(r - r_0))$ with r the distance from a fixed point. The layer at $x = x_0$ has width $O(1/\alpha)$; the values $\alpha = 20$ and $\alpha = 1000$ correspond to the mild and steep variants in [8]. We take $x_0 = 0.5$.

3 Parabolic Problems

Problem 2. Heat Equation

$$\boxed{u_t - D\,u_{xx} = f(x,t)}$$

Case 2a (manufactured). $(x,t) \in (0,L) \times (0,T]$, $u(0,t) = u(L,t) = 0$.

$$u(x,t) = A\,s(x)g(t),$$
$$f(x,t) = A\left[\beta t^{\beta-1} s(x) - D\,g(t)s''(x)\right].$$

Case 2b (free decay of the first eigenmode). $(x,t) \in (0,L) \times (0,T]$, $u(0,t) = u(L,t) = 0$, $u(x,0) = \sin(\pi x/L)$, and $f \equiv 0$. Separation of variables (see, e.g., [13, Ch. 4]) gives

$$u(x,t) = e^{-\lambda t}\sin(\pi x/L), \qquad \lambda = \frac{D\pi^2}{L^2}.$$

Case 2c (time-periodic source). $(x,t) \in (0,L) \times (0,T]$, $u(0,t) = u(L,t) = 0$, $u(x,0) = 0$, and, for a forcing frequency $\gamma > 0$,

$$f(x,t) = \sin(\pi x/L)\cos(\gamma t).$$

Since f is proportional to the first eigenfunction, the method of eigenfunction expansion (see, e.g., [4, Ch. 8]) reduces the problem to the scalar equation $a' + \lambda a = \cos(\gamma t)$, $a(0) = 0$, with $\lambda = D\pi^2/L^2$. The solution is

$$u(x,t) = a(t)\sin(\pi x/L), \qquad a(t) = \frac{\lambda\cos(\gamma t) + \gamma\sin(\gamma t) - \lambda e^{-\lambda t}}{\lambda^2 + \gamma^2}.$$

After the transient $e^{-\lambda t}$ has decayed, the solution oscillates with amplitude $(\lambda^2 + \gamma^2)^{-1/2}$ and a phase lag of $\arctan(\gamma/\lambda)$ relative to the source.

Problem 3. Convection–Diffusion Equation

$$\boxed{u_t + v\,u_x - D\,u_{xx} = f(x,t)}, \qquad v \in \mathbb{R}.$$

The Péclet number is $\mathrm{Pe} = vL/D$.

Case 3a (manufactured). $(x,t) \in (0,L) \times (0,T]$, $u(0,t) = u(L,t) = 0$.

$$u(x,t) = A\,s(x)g(t),$$
$$f(x,t) = A\left[\beta t^{\beta-1} s(x) + v\,g(t)s'(x) - D\,g(t)s''(x)\right].$$

Case 3b (steady boundary layer, no source; Patankar [10, Ch. 5]). $x \in (0,L)$, $u(0) = 0$, $u(L) = 1$, $f \equiv 0$, $v \neq 0$. The time-independent solution is

$$u(x) = \frac{e^{\mathrm{Pe}\,x/L} - 1}{e^{\mathrm{Pe}} - 1}.$$

For $v > 0$ and $\mathrm{Pe} \gg 1$, u is close to 0 except in a boundary layer of width $O(D/v)$ at $x = L$. This solution is the basis of the exponential differencing scheme in [10].

Case 3c (steady boundary layer, constant source). $x \in (0,L)$, $u(0) = u(L) = 0$, $f \equiv 1$, $v \neq 0$. Direct integration of $v\,u' - D\,u'' = 1$ gives

$$u(x) = \frac{1}{v}\left[x - L\,\frac{e^{\mathrm{Pe}\,x/L} - 1}{e^{\mathrm{Pe}} - 1}\right].$$

For $v > 0$ and $\mathrm{Pe} \gg 1$, $u \approx x/v$ away from $x = L$, with a boundary layer of width $O(D/v)$ at $x = L$. This is a standard model problem for singularly perturbed convection–diffusion equations; see [12].

Case 3d (advected Gaussian pulse). $x \in \mathbb{R}$, $f \equiv 0$. For $\sigma_0 > 0$ and $x_0 \in \mathbb{R}$,

$$u(x,t) = \frac{\sigma_0}{\sigma(t)} \exp\!\left(-\frac{(x - x_0 - vt)^2}{2\sigma(t)^2}\right), \qquad \sigma(t)^2 = \sigma_0^2 + 2Dt.$$

This follows from the fundamental solution of the heat equation [3, Sec. 2.3] and the change of variables $\xi = x - vt$. On a bounded interval, the Dirichlet data are taken from the exact solution; if the interval is chosen so that the pulse remains far from the boundary, these data are negligibly small.

Problem 4. Viscous Burgers Equation

The equation is quasilinear and, since $D > 0$, parabolic.

$$\boxed{u_t + u\,u_x - D\,u_{xx} = f(x,t)}$$

Case 4a (manufactured). $(x,t) \in (0,L) \times (0,T]$, $u(0,t) = u(L,t) = 0$.

$$u(x,t) = A\,s(x)g(t),$$
$$f(x,t) = A\left[\beta t^{\beta-1} s(x) - D\,g(t)s''(x)\right] + A^2 g(t)^2 s(x)s'(x).$$

Case 4b (sine initial data; Basdevant et al. [1]). $(x,t) \in (-1,1) \times (0,1]$, $u(\pm 1,t) = 0$, $u(x,0) = -\sin(\pi x)$, $f \equiv 0$, and $D = 0.01/\pi$. The Cole–Hopf transformation gives

$$u(x,t) = -\frac{\displaystyle\int_{-\infty}^{\infty} \sin\bigl(\pi(x-\eta)\bigr)\, h(x-\eta)\, e^{-\eta^2/(4Dt)}\, d\eta}{\displaystyle\int_{-\infty}^{\infty} h(x-\eta)\, e^{-\eta^2/(4Dt)}\, d\eta}, \qquad h(y) = \exp\!\left(-\frac{\cos(\pi y)}{2\pi D}\right).$$

With the substitution $\eta = 2\sqrt{Dt}\, z$, both integrals take the form $\int_{-\infty}^{\infty} e^{-z^2}(\cdot)\, dz$ and can be evaluated by Gauss–Hermite quadrature, as in [1]. The solution develops a steep internal layer at $x = 0$. This case was used as a benchmark for physics-informed neural networks by Raissi, Perdikaris, and Karniadakis [11].

Case 4c (viscous shock; Whitham [16, Ch. 4]). $x \in \mathbb{R}$, $f \equiv 0$. For constants $u_\ell > u_r$, the travelling wave

$$u(x,t) = \frac{u_\ell + u_r}{2} - \frac{u_\ell - u_r}{2} \tanh\!\left(\frac{(u_\ell - u_r)(x - ct)}{4D}\right), \qquad c = \frac{u_\ell + u_r}{2},$$

connects u_ℓ as $x \to -\infty$ to u_r as $x \to +\infty$ and moves with the Rankine–Hugoniot speed c. The shock width is $O\bigl(D/(u_\ell - u_r)\bigr)$. On a bounded interval, the Dirichlet data are taken from the exact solution.

4 Hyperbolic Problems

Problem 5. Two-Dimensional Wave Equation

$$\boxed{u_{tt} - D(u_{xx} + u_{yy}) = f(x,y,t)}, \qquad (x,y) \in \Omega = (0,L)^2, \quad t \in (0,T], \quad u = 0 \text{ on } \partial\Omega.$$

Define

$$S(x,y) = s(x)s(y), \qquad \Delta S = s''(x)s(y) + s(x)s''(y), \qquad \omega = \frac{\pi\sqrt{2D}}{L}.$$

Exact solution and initial conditions:

$$\begin{aligned} u(x,y,t) &= A\, S(x,y) \left[\cos(\omega t) + t^\beta\right], \\ u(x,y,0) &= A\, S(x,y), \\ u_t(x,y,0) &= 0. \end{aligned}$$

Source term:

$$f(x,y,t) = A\left[-\omega^2 \cos(\omega t) + \beta(\beta-1)t^{\beta-2}\right] S(x,y) - DA\left[\cos(\omega t) + t^\beta\right] \Delta S(x,y).$$

The initial condition $u_t(x,y,0) = 0$ requires $\beta > 1$. For $1 < \beta < 2$, the term $\beta(\beta-1)t^{\beta-2}$ is unbounded as $t \to 0^+$; we therefore take $\beta \geq 2$.

For the training profile, S is the first Dirichlet eigenfunction of $-\Delta$ on Ω, with eigenvalue $\lambda_1 = 2\pi^2/L^2$, and $\omega = \sqrt{D\lambda_1}$. Consequently, $A\, S \cos(\omega t)$ solves the homogeneous equation, and the source term reduces to

$$f(x,y,t) = A\left[\beta(\beta-1)t^{\beta-2} + \frac{2\pi^2 D}{L^2}\, t^\beta\right] S(x,y).$$

For the validation and reserved profiles, S is not an eigenfunction, and no such reduction occurs.

5 Time-Fractional Problems

The classification into elliptic, parabolic, and hyperbolic types is defined for local differential operators through their principal part. The Caputo derivative is a nonlocal operator in time, so this classification does not apply to the problems below. For $0 < \alpha < 1$, the problems in this section are time-fractional diffusion equations, also called subdiffusion equations; they reduce formally to the heat equation as $\alpha \to 1^-$. For $1 < \alpha < 2$, the corresponding equation is the diffusion-wave equation, which interpolates between the heat equation ($\alpha = 1$) and the wave equation ($\alpha = 2$); no such problem is included here.

Scalar Problem

Problem 6. Fractional Ordinary Differential Equation

$$\boxed{{}_0^C D_t^\alpha u(t) = f(t)}, \qquad t \in (0,1], \qquad u(0) = 0, \qquad 0 < \alpha < 1, \quad p > 0.$$

Exact solution and source term:

$$u(t) = t^p, \qquad f(t) = \frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha}.$$

The source term is bounded on $(0,1]$ if and only if $p \geq \alpha$.

Constant-Order Time-Fractional Diffusion

Problem 7. Time-Fractional Diffusion Equation

$$\boxed{{}_0^C D_t^\alpha u - D\, u_{xx} = f(x,t)}, \qquad 0 < \alpha < 1.$$

Exact solution and source term:

$$\begin{aligned} u(x,t) &= A\, s(x) g(t), \\ f(x,t) &= A \left[\frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha)}\, t^{\beta-\alpha} s(x) - D\, g(t) s''(x) \right]. \end{aligned}$$

The source term is bounded on $(0,L) \times (0,T]$ if and only if $\beta \geq \alpha$.

Problem 8. Multi-Term Time-Fractional Diffusion Equation

$$\boxed{\sum_{j=1}^{2} w_j\, {}_0^C D_t^{\alpha_j} u - D\, u_{xx} = f(x,t)}, \qquad 0 < \alpha_j < 1, \quad w_j > 0.$$

Exact solution and source term:

$$\begin{aligned} u(x,t) &= A\, s(x) g(t), \\ f(x,t) &= A \left[\sum_{j=1}^{2} w_j \frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha_j)}\, t^{\beta-\alpha_j} s(x) - D\, g(t) s''(x) \right]. \end{aligned}$$

The source term is bounded on $(0,L) \times (0,T]$ if and only if $\beta \geq \max(\alpha_1, \alpha_2)$.

Problem 9. Time-Fractional Diffusion Equation with Zero Initial Data

Problems 9–11 are posed on $(x,t) \in (0,\pi) \times (0,1]$ with $p > 0$, boundary conditions $u(0,t) = u(\pi,t) = 0$, and initial condition $u(x,0) = 0$.

$$\boxed{{}_0^C D_t^\alpha u = u_{xx} + f(x,t)}, \qquad 0 < \alpha < 1.$$

Exact solution and source term:

$$u(x,t) = t^p \sin x,$$
$$f(x,t) = \left[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha} + t^p \right] \sin x.$$

This problem coincides with Problem 7 for $D = 1$, $L = \pi$, $A = 1$, and the training profile, except that the temporal profile is t^p rather than $1 + t^\beta$.

Variable-Order Time-Fractional Diffusion

Problem 10. Time-Dependent Order

$$\boxed{{}_0^C D_t^{\alpha(t)} u = u_{xx} + f(x,t)}, \qquad \alpha(t) = c \sin^2 t, \qquad 0 < c < \frac{1}{\sin^2 1} \approx 1.412.$$

Exact solution and source term:

$$u(x,t) = t^p \sin x,$$
$$f(x,t) = \left[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(t))}\, t^{p-\alpha(t)} + t^p \right] \sin x.$$

Since $\sin^2 t$ is positive and increasing on $(0,1]$, the stated bound on c is equivalent to $0 < \alpha(t) < 1$ for all $t \in (0,1]$. Note that $\alpha(t) \to 0$ as $t \to 0^+$.

Problem 11. Space- and Time-Dependent Order

$$\boxed{{}_0^C D_t^{\alpha(x,t)} u = u_{xx} + f(x,t)}, \qquad \alpha(x,t) = c\, t^2 \sin x, \qquad 0 < c < 1.$$

Exact solution and source term:

$$u(x,t) = t^p \sin x,$$
$$f(x,t) = \left[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(x,t))}\, t^{p-\alpha(x,t)} + t^p \right] \sin x.$$

Since $\max_{(0,\pi)\times(0,1]} t^2 \sin x = 1$, attained at $(x,t) = (\pi/2, 1)$, the stated bound on c is equivalent to $0 < \alpha(x,t) < 1$ on $(0,\pi) \times (0,1]$. Note that $\alpha(x,t) \to 0$ as $t \to 0^+$ or as x approaches 0 or π.

6 Fourth-Order and Incompressible Flow Problems

Problem 12. Kuramoto–Sivashinsky Equation

$$\boxed{u_t + u\, u_x + u_{xx} + u_{xxxx} = f(x,t)}, \qquad x \in (0,L), \quad t \in (0,T].$$

Boundary conditions are periodic:

$$\frac{\partial^j u}{\partial x^j}(0,t) = \frac{\partial^j u}{\partial x^j}(L,t), \qquad j = 0,1,2,3, \quad t \in (0,T].$$

The term u_{xx} is destabilizing and u_{xxxx} is stabilizing. The Fourier mode e^{ikx} grows under the linear part if and only if $0 < |k| < 1$.

Case 12a (manufactured). Let $k = 2\pi/L$. Then

$$u(x,t) = A\sin(kx)\,g(t),$$
$$f(x,t) = A\left[\beta t^{\beta-1} + (k^4 - k^2)\,g(t)\right]\sin(kx) + \frac{A^2 k}{2}\,g(t)^2\sin(2kx),$$

with initial condition $u(x,0) = A\sin(kx)$. As in Cases 2a–4a, the source term is bounded if and only if $\beta \geq 1$.

Case 12b (chaotic dynamics; Kassam and Trefethen [6]). $L = 32\pi$, $T = 150$, $f \equiv 0$, and

$$u(x,0) = \cos\Big(\frac{x}{16}\Big)\left[1 + \sin\Big(\frac{x}{16}\Big)\right].$$

There is no closed-form solution. A reference solution is computed by a Fourier spectral method in space with the ETDRK4 scheme in time, as in [6]. The wavenumbers are $n/16$, $n \in \mathbb{Z}$, so the modes $1 \leq |n| \leq 15$ are linearly unstable. The solution evolves into spatiotemporal chaos.

Problem 13. Two-Dimensional Incompressible Navier–Stokes Equations

$$\boxed{\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p - \nu\,\Delta\mathbf{u} = \mathbf{f}, \qquad \nabla\cdot\mathbf{u} = 0},$$

where $\mathbf{u} = (u, v)$ is the velocity, p is the pressure (unit density), and $\nu > 0$ is the kinematic viscosity. The Reynolds number is $\mathrm{Re} = 1/\nu$.

Case 13a (Taylor–Green vortex; Taylor and Green [14]). $(x,y) \in \Omega = (0, 2\pi)^2$, $t \in (0, T]$, $\mathbf{f} \equiv \mathbf{0}$. Boundary conditions are periodic in both directions for $\mathbf{u}$ and p:

$$\mathbf{u}(0,y,t) = \mathbf{u}(2\pi,y,t), \qquad \mathbf{u}(x,0,t) = \mathbf{u}(x,2\pi,t),$$

and likewise for p and the first derivatives of $\mathbf{u}$. The initial condition is

$$\mathbf{u}(x,y,0) = (\cos x\,\sin y,\ -\sin x\,\cos y).$$

The exact solution is

$$u(x,y,t) = \cos x\,\sin y\,F(t),$$
$$v(x,y,t) = -\sin x\,\cos y\,F(t),$$
$$p(x,y,t) = -\frac{1}{4}\big[\cos(2x) + \cos(2y)\big]F(t)^2,$$
$$F(t) = e^{-2\nu t}.$$

The pressure is determined up to an additive constant, which is fixed by $\int_\Omega p\,dx\,dy = 0$; the exact pressure satisfies this condition. The pressure gradient balances the convective term exactly, and $\Delta\mathbf{u} = -2\mathbf{u}$. The flow therefore decays without changing shape. The benchmark also uses the vortex of integer wavenumber k, with $\cos x, \sin y, \ldots$ replaced by $\cos kx, \sin ky, \ldots$, $\cos 2x, \cos 2y$ by $\cos 2kx, \cos 2ky$, and $F(t) = e^{-2\nu k^2 t}$.

Case 13b (Kovasznay flow; Kovasznay [7]). $(x,y) \in \Omega = (-0.5,1) \times (-0.5,1.5)$, steady, $\mathbf{f} \equiv \mathbf{0}$. Let

$$\lambda = \frac{\mathrm{Re}}{2} - \sqrt{\frac{\mathrm{Re}^2}{4} + 4\pi^2}.$$

The exact solution is

$$\begin{aligned} u(x,y) &= 1 - e^{\lambda x}\cos(2\pi y), \\ v(x,y) &= \frac{\lambda}{2\pi}\, e^{\lambda x}\sin(2\pi y), \\ p(x,y) &= \frac{1}{2}\bigl(1 - e^{2\lambda x}\bigr). \end{aligned}$$

The velocity satisfies the Dirichlet condition $\mathbf{u} = (u,v)$ on $\partial\Omega$, with (u,v) given above. Since this velocity is divergence-free, the compatibility condition $\int_{\partial\Omega} \mathbf{u}\cdot\mathbf{n}\, ds = 0$ holds. No boundary condition is imposed on p. The pressure is determined up to an additive constant, which is fixed by matching the exact pressure at one point, e.g. $p(-0.5,-0.5) = \frac{1}{2}\bigl(1 - e^{-\lambda}\bigr)$. This solution models laminar flow behind a two-dimensional grid. A typical choice is $\mathrm{Re} = 40$.

7 Phase-Field and Electromagnetic Problems

Problem 14. Cahn–Hilliard Equation

$$\boxed{u_t = M\,\Delta\mu + f(x,y,t), \qquad \mu = u^3 - u - \varepsilon^2\Delta u}, \qquad (x,y) \in (0,L)^2, \quad t \in (0,T],$$

with mobility $M > 0$, interface parameter $\varepsilon > 0$, and periodic boundary conditions in both directions [2]. Without forcing the equation is the H^{-1} gradient flow of the free energy $\int \bigl[\frac{1}{4}(u^2-1)^2 + \frac{\varepsilon^2}{2}|\nabla u|^2\bigr]$, so $\int_\Omega u$ is conserved. Linearized about $u = 0$, the Fourier mode of wavenumber $|\mathbf{k}|$ grows at the rate $M|\mathbf{k}|^2(1 - \varepsilon^2|\mathbf{k}|^2)$, so the modes with $0 < |\mathbf{k}| < 1/\varepsilon$ are unstable.

Case 14a (manufactured). Let $k = 2\pi/L$, $c(x,y) = \cos(kx)\cos(ky)$, and $g(t) = 1 + t^\beta$. Then

$$\begin{aligned} u(x,y,t) &= A\, c(x,y)\, g(t), \\ f(x,y,t) &= A\,\beta t^{\beta-1} c(x,y) - M\left[3u^2\Delta u + 6u\,|\nabla u|^2 - \Delta u - 4\varepsilon^2 k^4 u\right], \end{aligned}$$

where $\Delta u = -2k^2 u$ and $|\nabla u|^2 = A^2 g(t)^2 k^2\bigl[\sin^2(kx)\cos^2(ky) + \cos^2(kx)\sin^2(ky)\bigr]$. The identity $\Delta(u^3) = 3u^2\Delta u + 6u|\nabla u|^2$ gives the nonlinear part. The source term is bounded if and only if $\beta \geq 1$.

Case 14b (spinodal decomposition). $L = 2\pi$, $M = 1$, $f \equiv 0$, and a small smooth perturbation of the mixed state,

$$u(x,y,0) = \sum_{j=1}^{5} a_j \cos\bigl(k_x^{(j)} x + k_y^{(j)} y + \phi_j\bigr),$$

with amplitudes $a_j \in [0.04, 0.1]$, integer wavenumbers, and phases that differ between the splits, as does $\varepsilon \in \{0.15, 0.12, 0.1\}$. The unstable modes grow until the field separates into regions with $u \approx \pm 1$ divided by interfaces of width $O(\varepsilon)$. There is no closed-form solution. A reference solution is computed by a Fourier spectral method with the ETDRK4 scheme [6], on 256^2 modes, and is checked against a 128^2 solve.

Problem 15. Maxwell's Equations (Two-Dimensional TE Mode)

In units in which the permittivity and permeability are equal and $c = 1/\sqrt{\epsilon\mu}$ is the wave speed, the time-domain Maxwell equations for the transverse electric field $\mathbf{E} = (E_x, E_y)$ and the out-of-plane magnetic field H_z read [5, 17]

$$\boxed{\mathbf{E}_t = c\,\nabla \times H_z - \mathbf{J}, \qquad (H_z)_t = -c\,\nabla \times \mathbf{E}},$$

with $\nabla \times H_z = (\partial_y H_z,\ -\partial_x H_z)$ and $\nabla \times \mathbf{E} = \partial_x E_y - \partial_y E_x$, on $\Omega = (0,a)\times(0,b)$ and $t \in (0,T]$. The boundary is a perfect electric conductor: the tangential electric field vanishes, $E_x = 0$ on $y = 0, b$ and $E_y = 0$ on $x = 0, a$.

Case 15a (resonant cavity mode). $\mathbf{J} \equiv \mathbf{0}$. For integers $m, n \geq 1$, let $\omega = c\pi\sqrt{(m/a)^2 + (n/b)^2}$. The standing wave

$$\begin{aligned} H_z &= \cos\frac{m\pi x}{a}\cos\frac{n\pi y}{b}\cos\omega t, \\ E_x &= -\frac{cn\pi}{b\omega}\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\sin\omega t, \\ E_y &= \frac{cm\pi}{a\omega}\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}\sin\omega t \end{aligned}$$

is the TE_{mn} eigenmode of the cavity, with $\mathbf{E}(x, y, 0) = \mathbf{0}$.

Case 15b (current-driven field; manufactured). For $A \neq 0$ and $\beta \geq 1$,

$$\begin{aligned} \mathbf{E} &= A\,t^{\beta}\Big(\cos\frac{\pi x}{a}\sin\frac{\pi y}{b},\ -\sin\frac{\pi x}{a}\cos\frac{\pi y}{b}\Big), \\ H_z &= cA\pi\Big(\frac{1}{a}+\frac{1}{b}\Big)\frac{t^{\beta+1}}{\beta+1}\cos\frac{\pi x}{a}\cos\frac{\pi y}{b}, \end{aligned}$$

which satisfies Faraday's law exactly and vanishes at $t = 0$. The current density is $\mathbf{J} = c\,\nabla\times H_z - \mathbf{E}_t$.

Problem 16. Time-Harmonic Electromagnetic Equation

For a field oscillating as $e^{-i\omega t}$ in a lossless medium, Maxwell's equations reduce to the curl–curl equation [9]

$$\boxed{\nabla \times \nabla \times \mathbf{E} - k^2\mathbf{E} = \mathbf{f}}, \qquad (x,y) \in \Omega = (0,1)^2, \qquad \mathbf{n}\times\mathbf{E} = 0 \text{ on } \partial\Omega,$$

with wavenumber $k > 0$ and, in two dimensions, $\nabla\times\nabla\times\mathbf{E} = (\partial_y w,\ -\partial_x w)$, $w = \partial_x E_y - \partial_y E_x$. The problem is uniquely solvable when k^2 is not a Maxwell eigenvalue of the square, $\pi^2(m^2+n^2)$ with integers $m, n \geq 0$ not both zero; every case takes k strictly between two consecutive eigenvalues.

Cases 16a and 16b (manufactured). For integers $p, n, m, q \geq 1$ and amplitudes A, B,

$$\begin{aligned} E_x &= A\cos(p\pi x)\sin(n\pi y), \qquad E_y = B\sin(m\pi x)\cos(q\pi y), \\ f_x &= \pi^2\big[n^2 A\cos(p\pi x)\sin(n\pi y) - mqB\cos(m\pi x)\sin(q\pi y)\big] - k^2 E_x, \\ f_y &= \pi^2\big[m^2 B\sin(m\pi x)\cos(q\pi y) - npA\sin(p\pi x)\cos(n\pi y)\big] - k^2 E_y. \end{aligned}$$

The tangential components vanish on the boundary. Case 16a uses a low frequency ($k < \pi$). Case 16b uses a high frequency near the resonances, for example $k = 10.6$, between $\pi\sqrt{10}$ and $\pi\sqrt{13}$, together with higher modes.

Numbers of the public cases

Training (evolution) and validation (development) cases. The literature cases use the values below. The manufactured cases (1a, 2a, 3a, 4a, 5, 7, 8) draw L, T, A, D, β (and v, α, w_j) per case and split from the ranges in `pdebench.manufactured`. The sealed test splits (test: 17 cases, test2 and test3: 20 cases each) use further values that are withheld here.

Case	Title	Training	Validation
1b	Poisson, multiscale	—	$D = 1$
1c	Poisson, interior layer	$D = 1$, $\alpha = 20$, $x_0 = 0.5$	—
2b	Heat, free decay	$L = 2$, $D = 0.3$	—
2c	Heat, time-periodic source	—	$T = 2$, $D = 0.2$, $\gamma = 2\pi$
3b	Convection–diffusion, steady boundary layer	$v = 1$, $D = 0.05$	—
3c	Convection–diffusion, steady layer with constant source	—	$v = 1$, $D = 0.02$
3d	Convection–diffusion, advected Gaussian	$L = 4$, $v = 1$, $D = 0.02$, $x_0 = 1$, $\sigma_0 = 0.2$	$L = 4$, v = -0.8, $D = 0.015$, $x_0 = 2.5$, $\sigma_0 = 0.15$
4b	Burgers, sine data	—	$D = 0.01/\pi$
4c	Burgers, viscous shock	$L = 1$, $u_\ell = 1$, $u_r = 0$, $D = 0.02$	—
6	Fractional ODE	$\alpha = 0.4$, $p = 2$	$\alpha = 0.7$, $p = 0.8$
9	Time-fractional diffusion, zero initial data	$\alpha = 0.5$, $p = 2$	$\alpha = 0.8$, $p = 1.5$
10	Variable order alpha(t)	$c = 0.8$, $p = 2$	$c = 1.3$, $p = 2.5$
11	Variable order alpha(x,t)	$c = 0.5$, $p = 2$	$c = 0.9$, $p = 3$

Numbers of the public cases, Problems 12–16

Case	Title	Training	Validation
12a	Kuramoto–Sivashinsky, manufactured	$L = 10.52$, $A = 1.465$, $\beta = 1.175$, $T = 0.5839$	$L = 15.19$, $A = 0.8677$, $\beta = 1.257$, $T = 0.9132$
12b	Kuramoto–Sivashinsky, chaotic (Kassam–Trefethen)	$L = 32\pi$, $T = 30$	$L = 32\pi$, $T = 60$
13a	Navier–Stokes, Taylor–Green vortex	$k = 1$, $\nu = 0.1$, $T = 1$	$k = 2$, $\nu = 0.05$, $T = 1$
13b	Navier–Stokes, Kovasznay flow	$\nu = 0.05$	$\nu = 0.025$
14a	Cahn–Hilliard, manufactured	$L = 3.263$, $A = 0.5148$, $\beta = 1.762$, $M = 1.382$, $\varepsilon = 0.06735$, $T = 0.2427$	$L = 2.771$, $A = 0.306$, $\beta = 1.976$, $M = 1.326$, $\varepsilon = 0.09144$, $T = 0.2588$
14b	Cahn–Hilliard, spinodal decomposition	$\varepsilon = 0.15$, $M = 1$, $T = 0.5$	$\varepsilon = 0.12$, $M = 1$, $T = 0.4$
15a	Maxwell TE_z, resonant cavity mode	$a = 1$, $b = 1$, $m = 1$, $n = 1$, $c = 1$, $T = 1$	$a = 1$, $b = 1.5$, $m = 2$, $n = 1$, $c = 1.5$, $T = 1$
15b	Maxwell TE_z, current-driven (manufactured)	$a = 1.992$, $b = 1.984$, $A = 0.8852$, $\beta = 1.292$, $c = 1.881$, $T = 0.6036$	$a = 1.503$, $b = 1.301$, $A = 0.878$, $\beta = 1.036$, $c = 1.897$, $T = 0.9697$
16a	Time-harmonic electromagnetic, low frequency	$A = 1$, $B = 0.5$, $p = 1$, $n = 1$, $m = 1$, $q = 1$, $k = 2$	$A = 0.8$, $B = 1.2$, $p = 2$, $n = 1$, $m = 1$, $q = 2$, $k = 1.5$
16b	Time-harmonic electromagnetic, high frequency	$A = 1$, $B = 0.7$, $p = 2$, $n = 3$, $m = 3$, $q = 1$, $k = 10.6$	$A = 0.9$, $B = 1.1$, $p = 1$, $n = 2$, $m = 2$, $q = 3$, $k = 7.6$

The sealed test split of Problems 12–16 has one case per row above (10 cases); its numbers are

withheld.

3 PDE Data Inventory suite

Twelve families from *PDE Data Inventory* (X. Kong, 19 October 2026), with inputs from the DiffusionPDE test set (Huang et al. 2024) where the inventory leaves gaps. The library adds the following classes to the ten classes of Problems 1–11:

$$\begin{aligned}
&\text{darcy_2d}: \quad -\nabla\cdot\big(a(x,y)\nabla u\big) = f(x,y)\\
&\text{poisson_2d}: \quad u_{xx}+u_{yy} = f(x,y)\\
&\text{helmholtz_2d}: \quad u_{xx}+u_{yy}+k^2u = f(x,y)\\
&\text{vorticity_ns}: \quad \omega_t+\mathbf{v}\cdot\nabla\omega = \nu\Delta\omega-\mu\omega+q, \quad \mathbf{v}=(\psi_y,-\psi_x),\ -\Delta\psi=\omega\\
&\text{allen_cahn_2d}: \quad u_t = \nabla\cdot(\kappa\nabla u)+\lambda(u-u^3)+f\\
&\text{burgers_2d}: \quad u_t+c_x uu_x+c_y uu_y = \nabla\cdot(\kappa\nabla u)+f\\
&\text{adr_2d}: \quad u_t+c_x uu_x+c_y uu_y = \nabla\cdot(\kappa\nabla u)+\lambda(u-u^3)+f\\
&\text{elliptic_4d}: \quad -\nabla\cdot(a\nabla u) = f \text{ in } B^3\times(-1,1), \quad u=0 \text{ on the boundary}
\end{aligned}$$

Family	Class	Layout	Limit	Tol.	Training	Validation
1-D Burgers, fixed Gaussian initial state	burgers	1d_periodic	90 s	0.002	$D = 0.1$	$D = 0.07$
1-D Burgers, random periodic initial state	burgers	1d_periodic	90 s	0.002	instance 0	instance 1
1-D linear convection–diffusion	convection_diffusion	1d_dirichlet	90 s	0.002	$x_0 = 5$, $\sigma_0 = 1$	$x_0 = 4$, $\sigma_0 = 0.8$
Darcy flow, binary permeability	darcy_2d	2d_steady	300 s	0.002	instance 0	instance 1
Poisson equation, random forcing	poisson_2d	2d_steady	300 s	0.002	instance 0	instance 1
Inhomogeneous Helmholtz, k = 1	helmholtz_2d	2d_steady	300 s	0.002	instance 0	instance 1
2-D Navier–Stokes vorticity, unit torus	vorticity_ns	2d_periodic_time	300 s	0.002	instance 0	instance 1
2-D Kolmogorov flow with drag	vorticity_ns	2d_periodic_time	300 s	0.002	own draw	own draw
Heterogeneous Allen–Cahn	allen_cahn_2d	2d_periodic_time	300 s	0.002	$\lambda = 0.840$	$\lambda = 0.833$
2-D stochastic scalar Burgers	burgers_2d	2d_periodic_time	300 s	0.002	$c_x = 0.813$, $c_y = 0.546$	$c_x = 1.020$, $c_y = 0.504$
Stochastic nonlinear advection–diffusion–reaction	adr_2d	2d_periodic_time	300 s	0.002	$c_x = 0.617$, $c_y = 0.255$, $\lambda = 0.520$	$c_x = 0.644$, $c_y = 0.378$, $\lambda = 0.375$
4-D elliptic, invariant coefficient	elliptic_4d	4d_points	300 s	0.05	own draw	own draw

The sealed splits `inventory_test` and `inventory_test2` (DiffusionPDE instances 2 and 3, and further draws) are withheld. References are computed at high resolution from the same continuous inputs: pseudospectral ETDRK4 (1-D Burgers), finite volumes on an $8\times$ refined grid (Darcy), sine series (Poisson, Helmholtz), integrating-factor RK4 (Navier–Stokes, Kolmogorov), pseudospectral RK4 (Allen–Cahn, 2-D Burgers, ADR), 63^4 finite volumes (4-D elliptic) and the closed form (linear convection–diffusion). Each is checked against a coarser solve; the tolerance is at least five times the reference error.

4 Manufactured suites (code protocol)

Generated from `src/pde_rsi/tasks.py` by `scripts/write_problem_latex.py`. Every problem has a manufactured exact solution; agents see the forcing and initial data only as black boxes.

4.1 One-dimensional benchmark (48 public, 24 reserved cases)

All cases use $x \in [0, L]$, $u(0,t) = u(L,t) = 0$ and the exact solution

$$u(x,t) = A\, s(x)\,\big(1 + t^{\beta}\big) \quad \text{(Poisson: } u(x) = A\, s(x)\text{)}, \qquad u(x,0) = A\, s(x),$$

with the spatial profile fixed by the split:

$$s(x) = \begin{cases} \sin(\pi x/L) & \text{train,} \\ 4\,\dfrac{x}{L}\Big(1 - \dfrac{x}{L}\Big) & \text{validation,} \\ \sin(\pi x/L) + \frac{1}{4}\sin(3\pi x/L) & \text{reserved.} \end{cases}$$

Write $g(t) = 1 + t^{\beta}$.

4.1.1 Poisson

$$-D\, u_{xx} = f(x), \quad x \in (0, L)$$
$$f = -D\, A\, s''(x)$$

Case	Split	L	D	A	Profile
`poisson-g0-v0`	train	1	0.05	0.3	sine
`poisson-g0-v1`	train	1.2	0.1	0.4	sine
`poisson-g0-v2`	train	1	0.15	0.5	sine
`poisson-g0-v3`	train	1.2	0.2	0.6	sine
`poisson-g1-v0`	validation	1	0.05	0.3	quadratic
`poisson-g1-v1`	validation	1.2	0.1	0.4	quadratic
`poisson-g1-v2`	validation	1	0.15	0.5	quadratic
`poisson-g1-v3`	validation	1.2	0.2	0.6	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.2 Heat

$$u_t - D\, u_{xx} = f(x,t), \quad x \in (0, L),\ t \in (0, T]$$
$$f = A\big[\beta t^{\beta-1} s(x) - D\, g(t)\, s''(x)\big]$$

Case	Split	L	T	D	A	β	Profile
`heat-g0-v0`	train	1	0.25	0.05	0.3	1	sine
`heat-g0-v1`	train	1.2	0.25	0.1	0.4	1	sine
`heat-g0-v2`	train	1	0.35	0.15	0.5	1	sine
`heat-g0-v3`	train	1.2	0.35	0.2	0.6	1	sine
`heat-g1-v0`	validation	1	0.25	0.05	0.3	2	quadratic
`heat-g1-v1`	validation	1.2	0.25	0.1	0.4	2	quadratic
`heat-g1-v2`	validation	1	0.35	0.15	0.5	2	quadratic
`heat-g1-v3`	validation	1.2	0.35	0.2	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.3 Convection–diffusion

$$u_t + v\,u_x - D\,u_{xx} = f(x,t)$$
$$f = A\big[\beta t^{\beta-1} s - D\,g\,s'' + v\,g\,s'\big]$$

Case	Split	L	T	D	v	A	β	Profile
`convection_diffusion-g0-v0`	train	1	0.25	0.05	0.25	0.3	1	sine
`convection_diffusion-g0-v1`	train	1.2	0.25	0.1	0.35	0.4	1	sine
`convection_diffusion-g0-v2`	train	1	0.35	0.15	0.45	0.5	1	sine
`convection_diffusion-g0-v3`	train	1.2	0.35	0.2	0.55	0.6	1	sine
`convection_diffusion-g1-v0`	validation	1	0.25	0.05	0.25	0.3	2	quadratic
`convection_diffusion-g1-v1`	validation	1.2	0.25	0.1	0.35	0.4	2	quadratic
`convection_diffusion-g1-v2`	validation	1	0.35	0.15	0.45	0.5	2	quadratic
`convection_diffusion-g1-v3`	validation	1.2	0.35	0.2	0.55	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.4 Viscous Burgers

$$u_t + u\,u_x - D\,u_{xx} = f(x,t)$$
$$f = A\big[\beta t^{\beta-1} s - D\,g\,s''\big] + A^2 g^2\,s\,s'$$

Case	Split	L	T	D	A	β	Profile
`burgers-g0-v0`	train	1	0.25	0.05	0.3	1	sine
`burgers-g0-v1`	train	1.2	0.25	0.1	0.4	1	sine
`burgers-g0-v2`	train	1	0.35	0.15	0.5	1	sine
`burgers-g0-v3`	train	1.2	0.35	0.2	0.6	1	sine
`burgers-g1-v0`	validation	1	0.25	0.05	0.3	2	quadratic
`burgers-g1-v1`	validation	1.2	0.25	0.1	0.4	2	quadratic
`burgers-g1-v2`	validation	1	0.35	0.15	0.5	2	quadratic
`burgers-g1-v3`	validation	1.2	0.35	0.2	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.5 Caputo subdiffusion

$$
{}_0^C D_t^{\alpha} u - D\, u_{xx} = f(x,t), \quad 0 < \alpha < 1
$$

$$
f = A\Big[\frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha)} t^{\beta-\alpha} s - D\, g\, s''\Big]
$$

Case	Split	L	T	D	A	β	α	Profile
`caputo-g0-v0`	train	1	0.25	0.05	0.3	2	0.4	sine
`caputo-g0-v1`	train	1.2	0.25	0.1	0.4	2	0.55	sine
`caputo-g0-v2`	train	1	0.35	0.15	0.5	2	0.7	sine
`caputo-g0-v3`	train	1.2	0.35	0.2	0.6	2	0.85	sine
`caputo-g1-v0`	validation	1	0.25	0.05	0.3	0.8	0.4	quadratic
`caputo-g1-v1`	validation	1.2	0.25	0.1	0.4	0.8	0.55	quadratic
`caputo-g1-v2`	validation	1	0.35	0.15	0.5	0.8	0.7	quadratic
`caputo-g1-v3`	validation	1.2	0.35	0.2	0.6	0.8	0.85	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.6 Multi-term Caputo diffusion

$$
\sum_{j=1}^{2} w_j\, {}_0^C D_t^{\alpha_j} u - D\, u_{xx} = f(x,t)
$$

$$
f = A\Big[\sum_j w_j \frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha_j)} t^{\beta-\alpha_j} s - D\, g\, s''\Big]
$$

Case	Split	L	T	D	A	β	(α_1, α_2)	(w_1, w_2)	Profile
`multi_term-g0-v0`	train	1	0.25	0.05	0.3	2	(0.3, 0.7)	(0.4, 1)	sine
`multi_term-g0-v1`	train	1.2	0.25	0.1	0.4	2	(0.35, 0.75)	(0.4, 1)	sine
`multi_term-g0-v2`	train	1	0.35	0.15	0.5	2	(0.4, 0.8)	(0.4, 1)	sine
`multi_term-g0-v3`	train	1.2	0.35	0.2	0.6	2	(0.45, 0.85)	(0.4, 1)	sine
`multi_term-g1-v0`	validation	1	0.25	0.05	0.3	0.8	(0.3, 0.7)	(0.4, 1)	quadratic
`multi_term-g1-v1`	validation	1.2	0.25	0.1	0.4	0.8	(0.35, 0.75)	(0.4, 1)	quadratic
`multi_term-g1-v2`	validation	1	0.35	0.15	0.5	0.8	(0.4, 0.8)	(0.4, 1)	quadratic
`multi_term-g1-v3`	validation	1.2	0.35	0.2	0.6	0.8	(0.45, 0.85)	(0.4, 1)	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

4.7 Two-dimensional wave equation (8 public, 4 reserved cases)

$$
\begin{aligned}
&u_{tt} - D\,(u_{xx} + u_{yy}) = f(x,y,t), \qquad (x,y) \in (0,L)^2,\ t \in (0,T],\\
&u = 0 \text{ on } \partial(0,L)^2, \qquad u(x,y,0) = A\, s(x) s(y), \qquad u_t(x,y,0) = 0,\\
&u(x,y,t) = A\, s(x)\, s(y)\, \big(\cos \omega t + t^{\beta}\big), \qquad \omega = \sqrt{D}\, \pi \sqrt{2}/L,\\
&f = A\Big[\big(-\omega^2 \cos \omega t + \beta(\beta-1) t^{\beta-2}\big) S - D\big(\cos \omega t + t^{\beta}\big) \Delta S\Big],\\
&S(x,y) = s(x)\, s(y),
\end{aligned}
$$

with s as in the 1D benchmark (train: sine, $\beta = 2.5$; validation: quadratic, $\beta = 1.5$, so $u_{tt} \sim t^{-1/2}$ is singular at $t = 0$; reserved: mixed, $\beta = 1.25$). $D = c^2$ is the squared wave speed. Frequencies ω: `wave2d-g0-v0`: 2.2214, `wave2d-g0-v1`: 2.7768, `wave2d-g0-v2`: 4.4429, `wave2d-g0-v3`: 4.6280, `wave2d-g1-v0`: 2.2214, `wave2d-g1-v1`: 2.7768, `wave2d-g1-v2`: 4.4429, `wave2d-g1-v3`: 4.6280.

Case	Split	L	T	$D = c^2$	A	β
`wave2d-g0-v0`	train	1	0.5	0.25	0.3	2.5
`wave2d-g0-v1`	train	1.2	0.5	0.5625	0.4	2.5
`wave2d-g0-v2`	train	1	0.75	1	0.5	2.5
`wave2d-g0-v3`	train	1.2	0.75	1.562	0.6	2.5
`wave2d-g1-v0`	validation	1	0.5	0.25	0.3	1.5
`wave2d-g1-v1`	validation	1.2	0.5	0.5625	0.4	1.5
`wave2d-g1-v2`	validation	1	0.75	1	0.5	1.5
`wave2d-g1-v3`	validation	1.2	0.75	1.562	0.6	1.5

Reserved cases: 4 (parameters kept evaluator-only).

4.8 Group FPDE examples (16 public, 4 reserved cases)

All start from rest and use $t \in (0, 1]$. The fractional PDEs use $x \in (0, \pi)$ with $u(0, t) = u(\pi, t) = 0$ and $u(x, 0) = 0$. The exact solutions are $u = t^p$ (FODE) and $u = t^p \sin x$ (FPDEs). The paper instances are $p = 3$ with order coefficient 0.5 (so $u = t^3$ or $u = t^3 \sin x$); the other cases vary the order and p.

4.9 Fractional ODE

$$ {}_0^C D_t^\alpha u(t) = f(t), \quad u(0) = 0, \qquad u(t) = t^p, \qquad f(t) = \frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha}. $$

Case	Split	α	p
`fode-g0-v0`	train	0.3	2.5
`fode-g0-v1`	train	0.7	3.5
`fode-g1-v0`	validation	0.5	3
`fode-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

4.10 Caputo FPDE

$$ {}_0^C D_t^\alpha u = u_{xx} + f(x, t), \qquad u = t^p \sin x, \qquad f(x, t) = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha} + t^p\Big] \sin x. $$

Case	Split	α	p
`fpde_caputo-g0-v0`	train	0.3	2.5
`fpde_caputo-g0-v1`	train	0.7	3.5
`fpde_caputo-g1-v0`	validation	0.5	3
`fpde_caputo-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

4.11 Time-variable-order FPDE

$$
{}_0^C D_t^{\alpha(t)} u = u_{xx} + f(x,t), \qquad \alpha(t) = c\,\sin^2 t, \qquad u = t^p \sin x,
$$
$$
f = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(t))}\, t^{p-\alpha(t)} + t^p\Big] \sin x.
$$

Case	Split	c	p
`fpde_variable_order-g0-v0`	train	0.3	2.5
`fpde_variable_order-g0-v1`	train	0.7	3.5
`fpde_variable_order-g1-v0`	validation	0.5	3
`fpde_variable_order-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

4.12 Space–time-variable-order FPDE

$$
{}_0^C D_t^{\alpha(x,t)} u = u_{xx} + f(x,t), \qquad \alpha(x,t) = c\,t^2 \sin x, \qquad u = t^p \sin x,
$$
$$
f = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(x,t))}\, t^{p-\alpha(x,t)} + t^p\Big] \sin x.
$$

Case	Split	c	p
`fpde_variable_order_xt-g0-v0`	train	0.3	2.5
`fpde_variable_order_xt-g0-v1`	train	0.7	3.5
`fpde_variable_order_xt-g1-v0`	validation	0.5	3
`fpde_variable_order_xt-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

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